GEC-CCP loss: calculation sheet

All residuals are computed after train-only plasma Z-score

1 Residual → Huber

$$e_t = \widehat{\tilde{y}}_t - \tilde{y}_t$$

$$H_1(e) = \tfrac{1}{2}e^2 \quad (|e| \leq 1)$$

$$H_1(e) = |e| - \tfrac{1}{2} \quad (|e| > 1)$$

2 Per-target spatial loss

$$\ell_t = L_{point} + 0.25 L_{boundary} + 0.10 L_{gradient} + 0.05 L_{multi}$$

point	boundary	gradient	multi
$\langle H_1(e) \rangle_{\Omega_p}$	$\langle H_1(e) \rangle_{D \leq 2\,px}$	$\mathrm{mean}_{r,z} H_1(\Delta e)$	$\mathrm{mean}_{s=2,4} H_1(A_s e)$

3 Equal weight by physical target family

$$L_{CCP} = \frac{1}{3}\left[\frac{\ell_{n_e} + \ell_{n_i}}{2} + \ell_{T_e} + \ell_{\phi}\right]$$

effective weights: $n_e = 1/6,\ n_i = 1/6,\ T_e = 1/3,\ \phi = 1/3$

casewise masked mean · plasma only · boundary band uses nearest chamber/part distance · Physics term disabled (weight 0)